

PREMISE:

Three tiny runs support costly large-scale decisions.

three random seeds



small models · many recipes



rank & make decision



expensive large model



Goal:

audit whether these tiny runs can be trusted to guide large decisions.

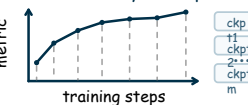
1 CHECKPOINT-NOISE PROXY

CAN ONE RUN CERTIFY CORRELATION?

Claim: a single run's last checkpoints can substitute for rerunning seeds and certify correlation 0.90

HOW WE AUDIT

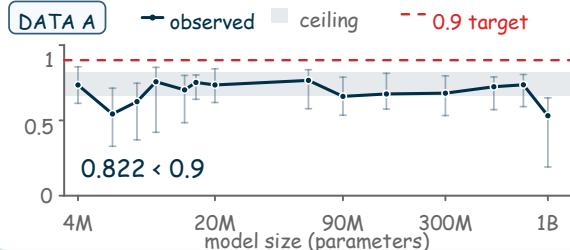
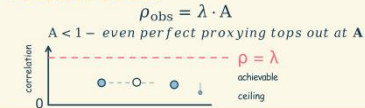
one run → many checkpoints



across seeds (same final ckpt)



CEILING LOGIC



COMPATIBLE WITH ITS CEILING BUT UNCERTIFIABLE

2 SOURCE EQUIVALENCE

CAN INIT-ONLY NOISE STAND IN?

Claim: an init-only noise floor can stand in for data-order noise.

HOW WE AUDIT

initialization (I)
(different seeds)



data order (D)
(different orders)



init-only arm (I)
many seeds,
fixed order

order-only arm (D)
many orders,
fixed init

σ_{init} = SD over
init-only arm
 σ_{order} = SD over
order-only arm

$$r = \sigma_{\text{init}} / \sigma_{\text{order}}$$

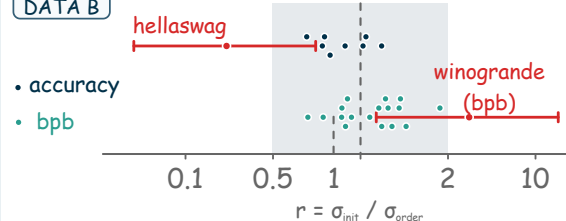
$$0.5 \leq r \leq 2$$

undecidable if
no CI fits

why single-source is not enough

Full model: $f(I, D) = \mu + a(I) + b(D) + c(I, D)$
Single-source arms estimate $a(I)$ or $b(D)$ only.
Interaction $c(I, D)$ is not identified.

DATA B



UNDECIDABLE

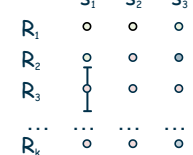
3 DECISION-INSTRUMENT SNR

DO SMALL-SCALE RANKINGS CARRY SIGNAL?

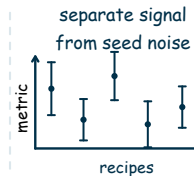
Claim: small-scale rankings carry enough signal over seed noise to support decisions.

HOW WE AUDIT

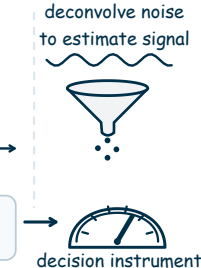
recipes × seeds



• recipe mean
I within-recipe SD(seed noise)

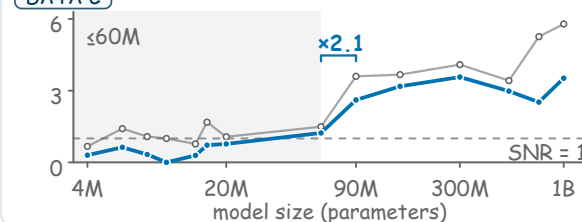


signal strength
among recipes



flat up to 60M, jumps $\times 2.1$ at 60M → 90M.

DATA C



NOISE-DOMINATED